

Tree Coin Collecting II

Time limit: 1.00 s

Memory limit: 512 MB

You are given a tree with n nodes. Some nodes contain a coin.

Your task is to answer q queries of the form: what is the shortest length of a path from node a to node b that visits all nodes with coins?

Input

The first line contains two integers n and q : the number of nodes and queries. The nodes are numbered $1, 2, \dots, n$.

The second line contains n integers $c_1, c_2, \dots, c_n$. If $c_i = 1$, node i has a coin. If $c_i = 0$, node i doesn't have a coin. You can assume at least one node has a coin.

Then there are $n - 1$ lines describing the edges. Each line contains two integers a and b : there is an edge between nodes a and b .

Finally, there are q lines describing the queries. Each line contains two integers a and b : the start and end nodes.

Output

Print q integers: the answers to the queries.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input	Output
5 4 1 0 0 1 0 2 4 2 3 1 3 3 5 1 5 3 2 4 4 5 5	6 5 6 8